



todays date: 4/05/2026

https://www.youtube.com/watch?v=Ubab6STQZLc&list=PL3eEXnCBViH_ePbZWc1nKZuyfru6sgioD&index=3

INSTRUCTIONS FOR GEMINI: i have placed some prompts in here for you starting with
`GEMINI:`

Finite Automata

finite automata represents a regular set

$FA = (Q, \Sigma, \delta, q_0, F)$

Σ = input alphabets

Q = set of states

q_0 = initial state

F = set of final states

δ = transition function

input (W) $\rightarrow$ FA (acceptor/recognizer)

if W belongs to L (valid strings), halts at final state

if w not belongs to L (invalid strings), halts at non final state

FA always halts

FA

$\wedge$

/ \

DFA NFA

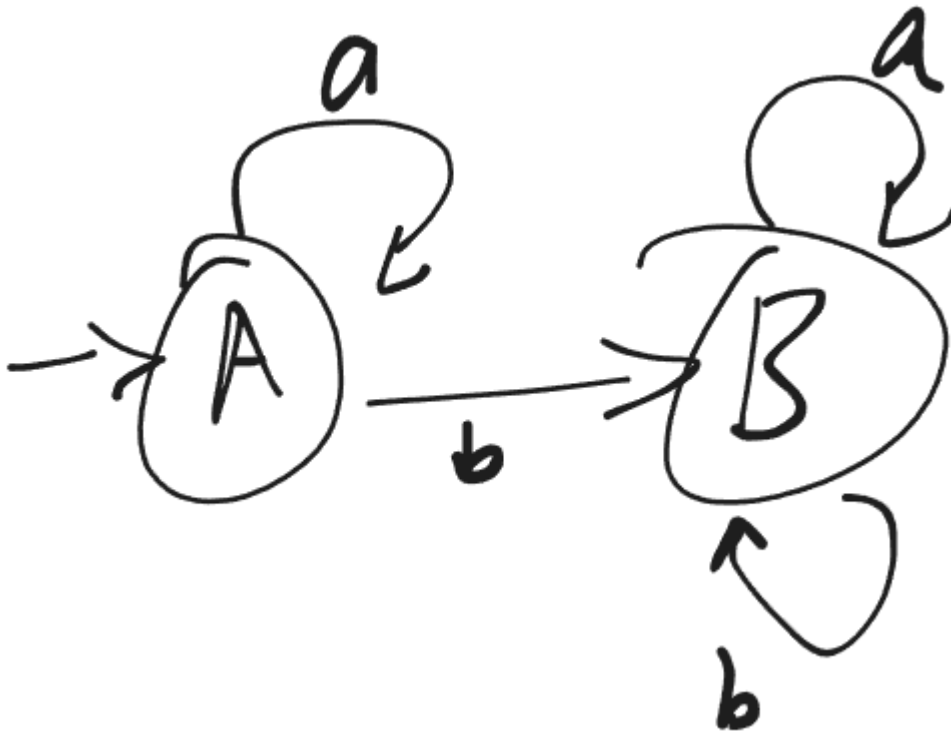
$\wedge$

/ \

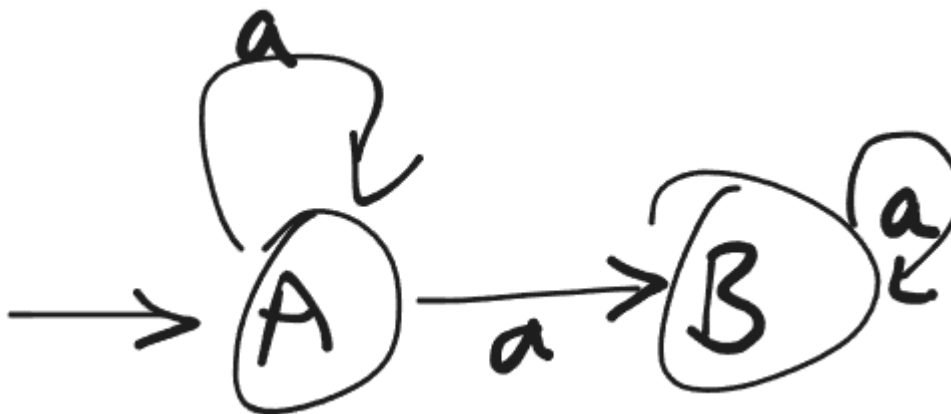
without E moves with E moves

DFA and NFA are equivalent

DFA = $Q \times E(\sigma)$



1.



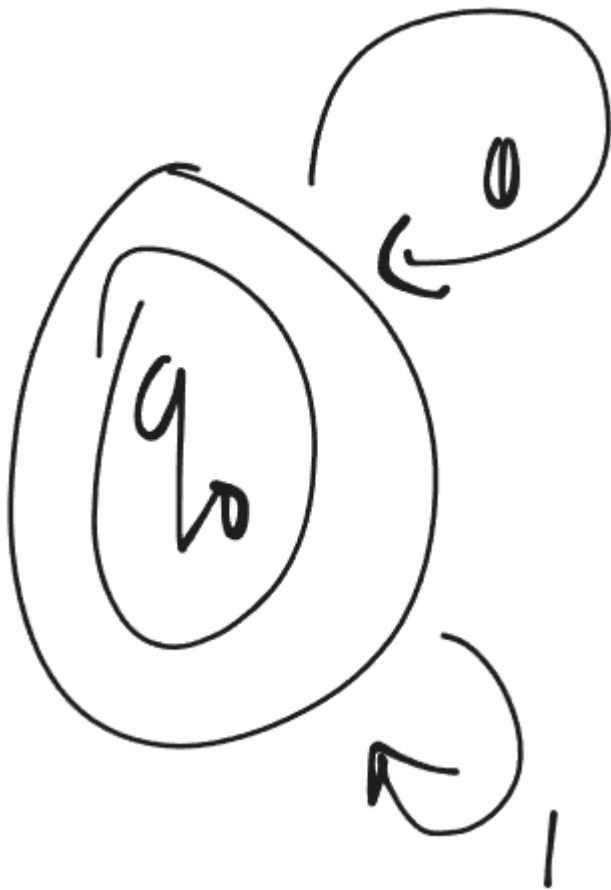
2.

1st is DFA, 2nd is not DFA GEMINI: explain

what is DFA?

$\delta = Q \times \sigma \rightarrow Q$

from every state in Q , for every input symbol in σ , there is exactly one transition to next state in Q

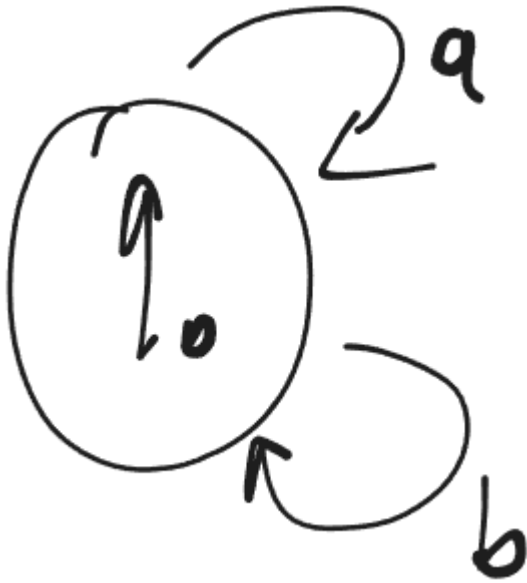


DFA for Set of all binary strings

if every state is final state in DFA, then the language is Σ^*

Empty set over $\Sigma = \{a,b\}$

$L = \{\}$



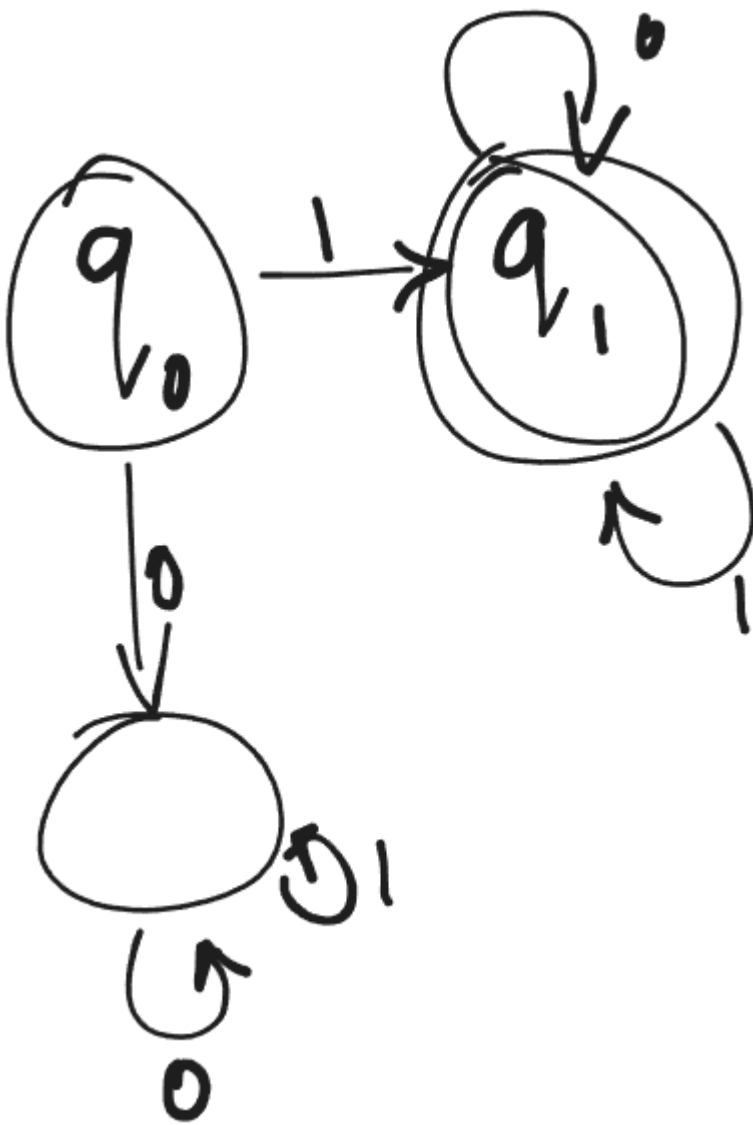
no path to a final state from this non final state
also called a dead state.

If every state is non final in DFA, then the language accepted by the DFA is $\{\}$ (Φ)

For every regular language accepted by only one (unique) DFA

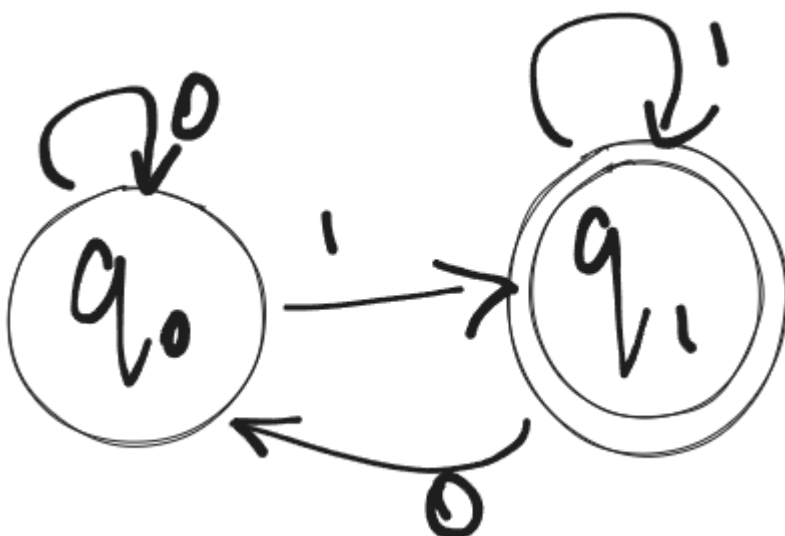
set of all binary strings starting with 1

$L = \{1...\}$

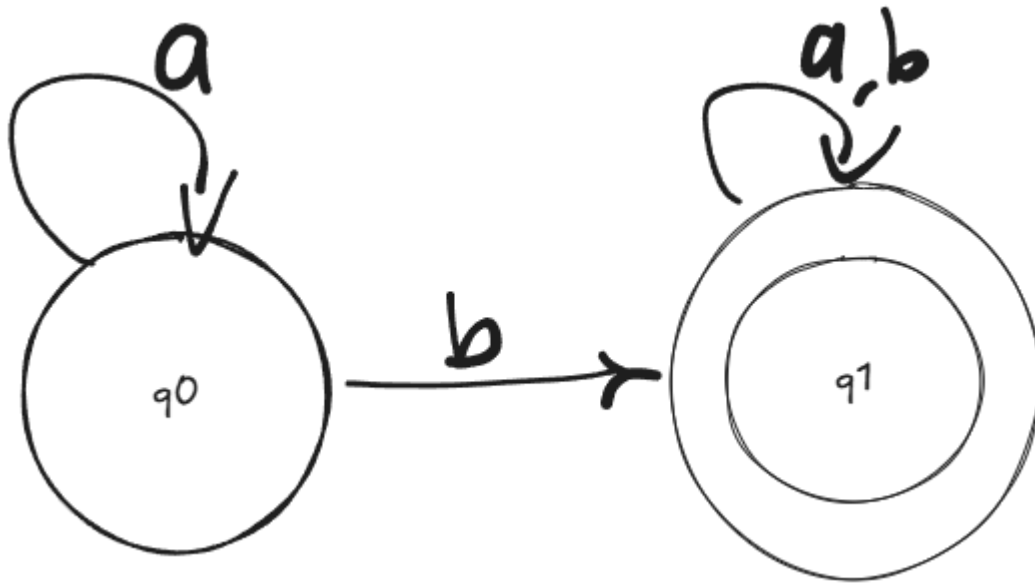


set of all binary strings ending with 1

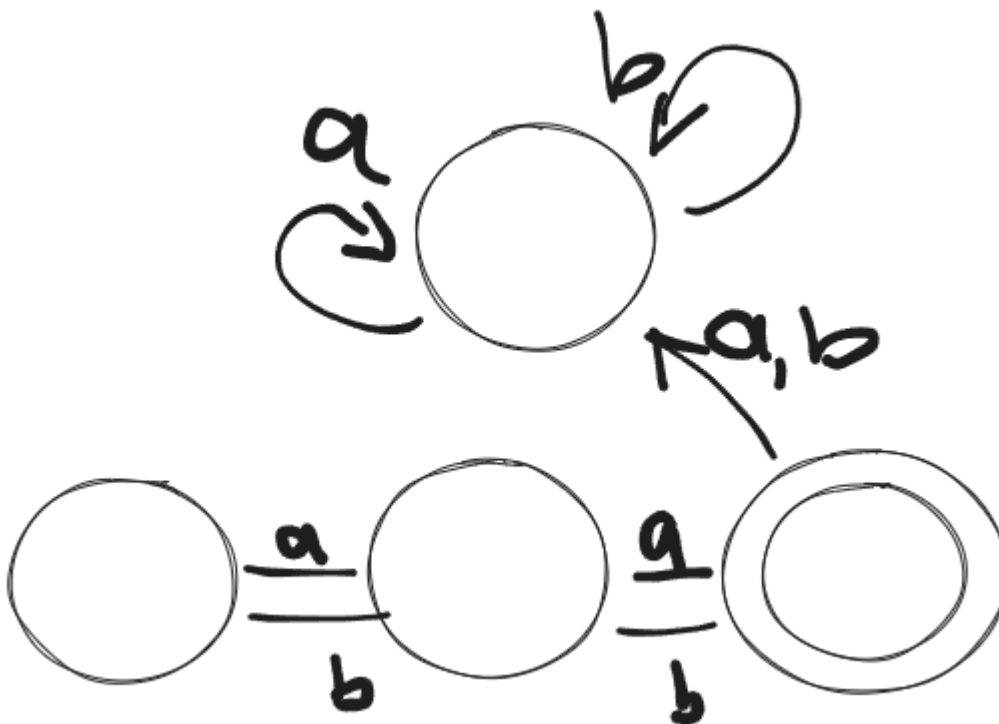
$L = \{...1\}$



set of all strings containing b, $\sigma = \{a,b\}$

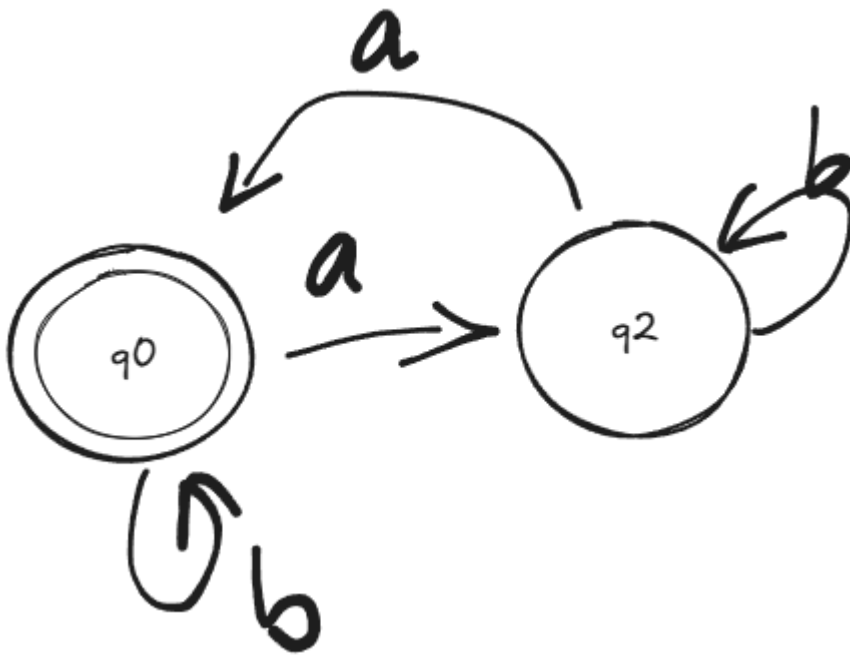


set of 2 length strings $\sigma = \{a,b\}$



$\sigma = \{a,b\}$

$L = \{w \mid w \text{ belongs } \{a,b\}^*, \#a(w) = \text{even}\}$



$L = \{w \mid w \text{ belongs to } \{a, b\}^*, w \text{ starts with } a, w \text{ contains } b\}.$

